```sql
CREATE DATABASE IF NOT EXISTS trigger_audit_db;

USE trigger_audit_db;

-- Create main table to store library books
CREATE TABLE library (
   library_id INT AUTO_INCREMENT PRIMARY KEY,
   title VARCHAR(255) NOT NULL
);

-- Create audit table to store history of updates and deletions
CREATE TABLE library_audit (
   audit_id INT AUTO_INCREMENT PRIMARY KEY,
   action_type ENUM('update', 'delete') NOT NULL,
   library_id INT NOT NULL,
   old_title VARCHAR(255),
   action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Trigger: Before Update (logs the old title before any modification)
DELIMITER //
CREATE TRIGGER before_library_update
BEFORE UPDATE ON library
FOR EACH ROW
BEGIN
   -- Insert old record data into audit table before updating
   INSERT INTO library_audit (action_type, library_id, old_title)
   VALUES ('update', OLD.library_id, OLD.title);
END //
DELIMITER ;

-- Trigger: Before Delete (logs the old title before deletion)
DELIMITER //
CREATE TRIGGER before_library_delete
BEFORE DELETE ON library
FOR EACH ROW
BEGIN
   -- Insert old record data into audit table before deletion
   INSERT INTO library_audit (action_type, library_id, old_title)
   VALUES ('delete', OLD.library_id, OLD.title);
END //
DELIMITER ;

-- Insert a sample record
INSERT INTO library (title)
VALUES ('to kill a mockingbird');

-- Update record to fire BEFORE UPDATE trigger
UPDATE library
SET title = '1984'
WHERE title = 'to kill a mockingbird';

-- Delete record to fire BEFORE DELETE trigger
DELETE FROM library
WHERE title = '1984';

-- Display log entries from the audit table
```

```
SELECT * FROM library_audit;

-- ------------------------------------------------------------
-- 📘 VIVA / REPORT COMMENTS
-- ------------------------------------------------------------
-- 🔹 TOPIC: Trigger-Based Audit System in MySQL
--
-- 🔹 WHAT IS A TRIGGER?
--    ▪ A Trigger in MySQL is a database object that is automatically
--      executed (or fired) when a specific event occurs on a table.
--    ▪ Events can be INSERT, UPDATE, or DELETE.
--    ▪ Triggers help in automating tasks such as validation, logging,
--      auditing, or enforcing business rules.
--
-- 🔹 TYPES OF TRIGGERS:
--    ▪ BEFORE Trigger → Executes before the action (used here).
--    ▪ AFTER Trigger  → Executes after the action is completed.
--
-- 🔹 PURPOSE OF THIS PROGRAM:
--    ▪ To automatically record all updates and deletions made to the
--      'library' table into the 'library_audit' table.
--
-- 🔹 WORKING:
--    1️⃣ When a book title is updated, BEFORE UPDATE trigger fires.
--    2️⃣ When a book record is deleted, BEFORE DELETE trigger fires.
--    3️⃣ The triggers insert the old title and its details into 'library_audit'.
--
-- 🔹 IMPORTANT KEYWORDS:
--    ▪ OLD → Refers to the existing data before the change.
--    ▪ ENUM → Restricts action type to fixed values ('update', 'delete').
--    ▪ TIMESTAMP → Automatically records the date and time of change.
--
-- 🔹 ADVANTAGES OF USING TRIGGERS:
--    ▪ Maintains automatic history of changes (auditing).
--    ▪ Improves data security and accountability.
--    ▪ Reduces manual coding of repetitive tasks.
--
-- 🔹 REAL-LIFE APPLICATIONS:
--    ▪ Banking systems (track balance updates)
--    ▪ Library management (track issued/returned books)
--    ▪ Employee systems (track salary or promotion updates)
--
-- 🔹 EXPECTED OUTPUT (for demonstration):
--    +----------+-------------+------------+----------------------+---------------------+
--    | audit_id | action_type | library_id | old_title            | action_time         |
--    +----------+-------------+------------+----------------------+---------------------+
--    |        1 | update      |          1 | to kill a mockingbird | 2025-09-02 11:23:19 |
--    |        2 | delete      |          1 | 1984                 | 2025-09-02 11:23:24 |
--    +----------+-------------+------------+----------------------+---------------------+
--
-- 🔹 VIVA TIP (Short Answer):
--    "A trigger is an automatic piece of code that runs in response to
--     events like INSERT, UPDATE, or DELETE on a table. In this example,
--     the trigger helps maintain an audit log of all book changes in the
--     library table without needing manual effort."
```

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